

Security, Sovereignty & Failure Research

Research objective

- Determine whether the proposed system can actually provide technical sovereignty and controlled agent execution, rather than merely claiming to be private.
- The objective is to discover **how the sovereignty claim can be falsified**, then determine how the system could technically prove it.

Research areas

Data sovereignty

- Air-gapped AI
- On-premise AI security
- Data sovereignty
- Local inference security
- Zero external data transfer
- Network isolation
- Network egress control
- External API prevention
- DNS isolation
- Firewall enforcement
- Network monitoring

Agent security

- Agent permissions
- Least privilege
- Tool authorization
- Filesystem restrictions
- Process restrictions
- Capability restrictions
- Agent audit trails
- Agent isolation

Code execution security

- Untrusted code execution
- Code sandboxing
- Container isolation
- Filesystem isolation

- Network isolation
- Resource limits
- Process isolation
- Sandbox escape
- Malicious generated code

Adversarial research

R11 specifically identifies threats including prompt manipulation, evaluation gaming, sandbox escape, malicious code, prompt injection and agent manipulation.

Research:

- Prompt injection
- Indirect prompt injection
- Malicious documents
- Tool manipulation
- Agent manipulation
- Excessive agency
- Privilege escalation
- Sandbox escape
- Malicious code
- Data exfiltration
- Credential exposure
- Resource exhaustion
- Denial of service
- Untrusted tool outputs

Keywords

R8 — Regulatory / Security Research

Sovereignty

- sovereign AI security architecture
- air-gapped AI security
- on-premise AI security
- AI data sovereignty
- confidential enterprise AI
- local AI data protection
- zero external data transfer AI
- offline AI security controls
- AI network isolation
- AI egress control
- network-isolated AI workloads

Network Proof

- network egress monitoring
- zero-egress architecture
- Linux network isolation
- application network isolation
- firewall egress filtering
- packet-level network monitoring
- AI network traffic auditing
- DNS egress monitoring
- network isolation verification
- air-gap verification techniques

Agent Security

- AI agent security
- local agent security
- agent permission model
- least privilege AI agents
- agent tool authorization
- agent capability restrictions
- agent filesystem isolation
- agent execution isolation
- agent audit trail
- agent security policy enforcement

Code Sandbox

- AI code execution sandbox security
- untrusted code execution
- sandboxed Python execution
- containerized code sandbox
- microVM code sandbox
- filesystem isolation for code execution
- network isolation for code sandbox
- CPU memory limits sandbox
- process isolation AI agents
- sandbox escape prevention
- malicious AI-generated code containment

R5 — Failure Research

Local AI Failure

- local LLM failure modes
- local VLM failure modes
- quantized model failure modes
- LLM hallucination in enterprise documents
- enterprise RAG failure modes
- RAG retrieval failures
- document OCR failure modes
- engineering drawing AI limitations
- multimodal AI failure modes
- local inference reliability problems

Agent Failure

- AI agent failure modes
- agent planning failures
- agent tool selection failures
- agent execution failures
- agent infinite loops
- agent retry failure
- agent hallucinated tool calls
- agent incorrect task completion
- agent verification failures
- autonomous agent reliability

Enterprise Workflow Failure

- AI-generated artifact errors
- AI approval note errors
- AI engineering document errors
- AI coding agent failures
- code generation verification failures
- AI calculation errors
- enterprise AI grounding failures
- incorrect internal knowledge retrieval
- stale enterprise knowledge retrieval

R6 — Innovation Research

- local multi-model agent architecture
- adaptive local model routing
- capability-aware model routing
- resource-aware AI model selection
- multimodal agentic RAG
- agentic document intelligence
- local multimodal enterprise agents

- air-gapped agentic AI
- sovereign agent architecture
- local AI execution environments
- autonomous enterprise document workflows
- AI agents with verifiable execution
- provenance-aware AI agents
- verifiable AI artifact generation
- self-verifying AI agents
- resource-aware agent orchestration
- dynamic model selection under GPU constraints
- local model collaboration
- heterogeneous model agent systems
- hybrid small-model large-model local agents
- model-specialized agent workflows
- local AI workflow verification
- AI execution provenance
- AI sovereignty verification mechanisms
- novel zero-egress AI architectures

Required proof model

For every security claim:

Claim

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Threat

↓

Attack mechanism

↓

Control

↓

Verification mechanism

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Residual risk

Example:

Claim:

"No confidential data leaves the system."

Threat:

Agent/application attempts external connection.

Control:

Network egress restriction.

Verification:

Network monitoring + connection logs.

Residual risk:

Undiscovered communication path.